ALGEBRA

बीजगणित

PRO LEVEL SHEET



ALGEBRA/(बीजगणित)

Pro Level Sheet

1. If $x^4+\frac{1}{x^4}=1154$, $x>0$, find $(2x-4\sqrt{2})^2$?

यदि $x^4+\frac{1}{x^4}=1154$, $x>0$, तो $(2x-4\sqrt{2})^2$?

(a) 25 (b) 36
(c) 16 (d) None of these

2. If $x>0$, and $x^4+\frac{1}{x^4}=2207$, what is the value of $x^7+\frac{1}{x^7}$?

यदि $x>0$, और $x^4+\frac{1}{x^4}=2207$, है तो $x^7+\frac{1}{x^7}$ का मान क्या होगा?

(a) 710649 (b) 710647
(c) 710654 (d) 710661

3. If $\left(x-\frac{1}{x}\right)=2\sqrt{2}$, & $x>1$, what is the value of $\left(x^6-\frac{1}{x^6}\right)$.

यदि $\left(x-\frac{1}{x}\right)=2\sqrt{2}$ और $x>1$ है, तो $\left(x^6-\frac{1}{x^6}\right)$ का मान क्या होगा?

(a) $372\sqrt{6}$ (b) $384\sqrt{6}$
(c) $396\sqrt{6}$ (d) $420\sqrt{6}$

4. If $x=\frac{5+\sqrt{21}}{2}$, find $x^{11}+x^7-2525x^6-527x^3$

यदि $x=\frac{5+\sqrt{21}}{2}$ है, तो $x^{11}+x^7-2525x^6-527x^3$ का मान ज्ञात करें?

(a) 5 (b) -5
(c) $\sqrt{5}$ (d) $-\sqrt{5}$

5. If $\frac{7(2\sqrt{2}-x)}{x}=7+\frac{14}{x^2}$ then $\sqrt{(23x^{64}+42x^{12}+163)}$

यदि $\frac{7(2\sqrt{2}-x)}{x}=7+\frac{14}{x^2}$ है, तो $\sqrt{(23x^{64}+42x^{12}+163)}$ का मान ज्ञात करें?

(a) 12 (b) $12\sqrt{2}$
(c) $12\sqrt{3}$ (d) 24

6. If $x^2-3\sqrt{7}x+13=0$, then $\left(x-\sqrt{7}\right)^6-\frac{1}{\left(x-\sqrt{7}\right)^6}$

यदि $x^2-3\sqrt{7}x+13=0$, तो $\left(x-\sqrt{7}\right)^6-\frac{1}{\left(x-\sqrt{7}\right)^6}$

(a) $80\sqrt{77}$ (b) $-80\sqrt{77}$
(c) $\sqrt{77}$ (d) None

7. If $x=\frac{\sqrt{3}+1}{\sqrt{3}-1}$ and $y=\frac{1}{x}$. Find the value of $\frac{x^3+\frac{1}{x^3}+8xy}{x^3-y^3}$.

यदि $x=\frac{\sqrt{3}+1}{\sqrt{3}-1}$ और $y=\frac{1}{x}$ तो $\frac{x^3+\frac{1}{x^3}+8xy}{x^3-y^3}$ का मान ज्ञात करें?

(a) $\frac{26\sqrt{3}}{45}$ (b) $\frac{2\sqrt{3}}{3}$
(c) $2\sqrt{3}$ (d) 1

8. If $a+b+c=4$ and $a^2+b^2+c^2=14$, $\frac{1}{a}+\frac{1}{b}+\frac{1}{c}=1$, find the value of abc.

यदि $a+b+c=4$ और $a^2+b^2+c^2=14$, $\frac{1}{a}+\frac{1}{b}+\frac{1}{c}=1$ हो, तो abc का मान ज्ञात कीजिए।

(a) $\frac{3}{2}$ (b) 2
(c) 1 (d) 0

9. If $x+y+z=-7$, $x^2+y^2+z^2=133$ & $xz=y^2$ then $z-x=?$ $(x<z<y)$

यदि $x+y+z=-7$, $x^2+y^2+z^2=133$ और $xz=y^2$, तो $z-x=?$ $(x<z<y)$ का मान क्या है?

(a) 4 (b) 5
(c) 6 (d) 7

10. If $(a+b+c)=17$, and $(a^2+b^2+c^2)=101$, find the value of $(a-b)^2+(b-c)^2+(c-a)^2$.

यदि $(a+b+c)=17$ और $(a^2+b^2+c^2)=101$ है, तो $(a-b)^2+(b-c)^2+(c-a)^2$ का मान क्या होगा?

(a) 12 (b) 14
(c) 10 (d) 16

11. If $(11x+8)^3+(2-x)^3-27(2x-5)^3=9(11x+8)(x-2)(2x-5)$. then what is the value of $\sqrt{x^2-36}$?

यदि $(11x+8)^3+(2-x)^3-27(2x-5)^3=9(11x+8)(x-2)(2x-5)$ है, तो $\sqrt{x^2-36}$ मान क्या होगा?

(a) $\frac{9}{2}$ (b) $\frac{7}{2}$
(c) $\frac{9}{4}$ (d) $\frac{7}{4}$

12. If $x^2-11x+1=0$, what is the value of $x^8-14159x^4+11$?

यदि $x^2-11x+1=0$ है, तो $x^8-14159x^4+11$ का मान क्या होगा?

(a) 9 (b) 10
(c) 12 (d) 11

13. If $\frac{2x}{x+2}+\frac{1}{2x+4}-\frac{12-x}{2x^2+4x}=a+\frac{b}{x}$ where a and b are integers, then the value of ab = ____.

यदि $\frac{2x}{x+2}+\frac{1}{2x+4}-\frac{12-x}{2x^2+4x}=a+\frac{b}{x}$ है, जहाँ a और b एक पूर्णांक है, तो ab बराबर होगा?

(a) 3 (b) – 6
(c) 6 (d) – 3

14. If $\frac{A}{L}+\frac{M}{B}=1$ & $\frac{B}{M}+\frac{N}{C}=1$, then the value of $\frac{L}{A}+\frac{C}{N}$ is:

यदि $\frac{A}{L}+\frac{M}{B}=1$ और $\frac{B}{M}+\frac{N}{C}=1$ है, तो $\frac{L}{A}+\frac{C}{N}$ का मान क्या है?

(a) 1 (b) $\frac{B}{M}$
(c) $\frac{M}{B}$ (d) 0

15. What is $\frac{x^6-24x^4+144x^2}{\left(x^2+4\sqrt{3}x+12\right)\left(x-2\sqrt{3}\right)^2}$ equal to?

$\frac{x^6-24x^4+144x^2}{\left(x^2+4\sqrt{3}x+12\right)\left(x-2\sqrt{3}\right)^2}$ के बराबर है?

(a) x^2 (b) x^2-2
(c) $x^2+2\sqrt{3}$ (d) $x^2-2\sqrt{3}$

16. If $p=\frac{\sqrt{3q+2}+\sqrt{3q-2}}{\sqrt{3q+2}-\sqrt{3q-2}}$ then what is the value of $p^2 - 3pq + 2$?

यदि $p=\frac{\sqrt{3q+2}+\sqrt{3q-2}}{\sqrt{3q+2}-\sqrt{3q-2}}$ है, तो $p^2 - 3pq + 2$ का मान है?

(a) 0 (b) 1
(c) 2 (d) 3

17. If $a^4+\frac{1}{a^4}=50$, $a > 0$ then the value of $a^3+\frac{1}{a^3}$.

यदि $a^4+\frac{1}{a^4}=50$ है, जहाँ $a > 0$ तो $a^3+\frac{1}{a^3}$ का मान है?

(a) $\sqrt{2\left(1+\sqrt{13}\right)}\left(-1+2\sqrt{13}\right)$
(b) $\sqrt{2\left(1-\sqrt{13}\right)}\left(-1+2\sqrt{13}\right)$
(c) $\sqrt{2\left(1+\sqrt{13}\right)}\left(-1-2\sqrt{13}\right)$
(d) $\sqrt{2\left(1+\sqrt{13}\right)}+\left(-1+2\sqrt{13}\right)$

18. If $\frac{6x}{2x^2+5x-2}=1$, $x > 0$, then the value of $x^3+\frac{1}{x^3}$ is:

यदि $\frac{6x}{2x^2+5x-2}=1$, $x > 0$ है, तो $x^3+\frac{1}{x^3}$ का मान है।

(a) $\frac{3}{8}\sqrt{17}$ (b) $\frac{5}{8}\sqrt{17}$
(c) $\frac{5}{16}\sqrt{17}$ (d) $\frac{3}{4}\sqrt{17}$

19. If $\left(x+\frac{1}{x}\right)=\sqrt{6}$ and $x > 1$, what is the value of $x^8-\frac{1}{x^8}$?

यदि $\left(x+\frac{1}{x}\right)=\sqrt{6}$ और $x > 1$, है, तो $x^8-\frac{1}{x^8}$ का मान ज्ञात कीजिए।

(a) $120\sqrt{3}$ (b) $128\sqrt{3}$
(c) $112\sqrt{3}$ (d) $108\sqrt{3}$

20. Which of the following statement is correct?

I. The value of $100^2 - 99^2 + 98^2 - 97^2 + 96^2 - 95^2 + 94^2 - 93^2 + ... + 22^2 - 21^2$ is 4840.

II. The value of $\left(k^2+\frac{1}{k^2}\right)\left(k-\frac{1}{k}\right)\left(k^4+\frac{1}{k^4}\right)\left(k+\frac{1}{k}\right)\left(k^4-\frac{1}{k^4}\right)$ is $k^{16}-\frac{1}{k^{16}}$

निम्नलिखित में से कौन सा कथन सही है?

I. $100^2 - 99^2 + 98^2 - 97^2 + 96^2 - 95^2 + 94^2 - 93^2 + ... + 22^2 - 21^2$ का मान 4840 है।

II. $\left(k^2+\frac{1}{k^2}\right)\left(k-\frac{1}{k}\right)\left(k^4+\frac{1}{k^4}\right)\left(k+\frac{1}{k}\right)\left(k^4-\frac{1}{k^4}\right)$ का मान $k^{16}-\frac{1}{k^{16}}$ हैं।

(a) Neither I nor II
(b) Only II
(c) Only I
(d) Both I and II

21. If $3a + 2b = 27$ and $27a^3 + 8b^3 = 1458$, then find 2ab.

यदि $3a + 2b = 27$ और $27a^3 + 8b^3 = 1458$ है, तो 2ab का मान ज्ञात कीजिए।

(a) 72 (b) 70
(c) 77 (d) 75

22. If $2a - b = 3$ and $8a^3 - b^3 = 999$, then the value of $4a^2 - b^2$.

यदि $2a - b = 3$ और $8a^3 - b^3 = 999$ है, तो $4a^2 - b^2$ का मान ज्ञात कीजिए।

(a) 63 (b) 65
(c) 61 (d) 67

23. If $(4x+2y)^3 + (4x-2y)^3 = 16\ (Ax^3 + Bxy^2)$, then what is the value of $\frac{1}{2}\left(\sqrt{A^2+B^2}\right)$?

यदि $(4x + 2y)^3 + (4x - 2y)^3 = 16\ (Ax^3 + Bxy^2)$ है, तो $\frac{1}{2}\left(\sqrt{A^2+B^2}\right)$ का मान ज्ञात करें।

(a) 8 (b) 3
(c) 5 (d) 7

24. If $(a + b + c) = 20$, and $a^2 + b^2 + c^2 = 152$, find the value of $a^3 + b^3 + c^3 - 3abc$.

यदि $(a + b + c) = 20$ और $a^2 + b^2 + c^2 = 152$ है, तो $a^3 + b^3 + c^3 - 3abc$ का मान ज्ञात कीजिए।

(a) 560 (b) 640
(c) 480 (d) 720

25. If $x = \sqrt{1+\frac{\sqrt{3}}{2}} - \sqrt{1-\frac{\sqrt{3}}{2}}$, then the value of $\frac{\sqrt{2}-x}{\sqrt{2}+x}$ will be closest to:

यदि $x = \sqrt{1+\frac{\sqrt{3}}{2}} - \sqrt{1-\frac{\sqrt{3}}{2}}$ है, तो $\frac{\sqrt{2}-x}{\sqrt{2}+x}$ का मान निकटतम होगा:

(a) 0.17 (b) 0.12
(c) 1.4 (d) 1.2

26. If $\frac{3\sqrt{5}-5}{3\sqrt{5}+5} = a+b\sqrt{5}$ then what is the value of b?

यदि $\frac{3\sqrt{5}-5}{3\sqrt{5}+5} = a+b\sqrt{5}$ तो b का मान क्या होगा?

(a) $\frac{7}{2}$ (b) $\frac{3\sqrt{5}}{2}$
(c) $\frac{3}{2}$ (d) $-\frac{3}{2}$

27. Simplify the given expression.
दिए गए व्यंजक को सरल कीजिए।

$$\left[(x-y)+\frac{y^2}{x+y}\right] \div \left[x^2+y^2+\frac{y^4}{x^2-y^2}\right] \times \frac{y^2}{x-y}$$

(a) $\frac{x^2}{y^2}$ (b) $\frac{x^2-y^2}{x^2y^2}$
(c) $\frac{y^2}{x^2}$ (d) x^2

28. If $a^3 - b^3 = 625$, $a^2 - b^2 = 25$ and $a + b = 5$ then find the value of $a^2 + ab + b^2$?
यदि $a^3 - b^3 = 625$, $a^2 - b^2 = 25$ और $a + b = 5$ है तो $a^2 + ab + b^2$ का मान ज्ञात कीजिये?
(a) 125 (b) 25
(c) 5 (d) 150

29. If $3a = 4b = 6c$ and $a + b + c = 27\sqrt{29}$, then what is the value of $\sqrt{a^2+b^2+c^2}$?

यदि $3a = 4b = 6c$ और $a + b + c = 27\sqrt{29}$, तो $\sqrt{a^2+b^2+c^2}$ का मान क्या होगा?

(a) $3\sqrt{81}$ (b) 81
(c) 85 (d) 87

30. If 2 is one root of equation $x^2 - px + 6 = 0$ and quadratic equation $x^2 + 2px + q = 0$ has equal roots then what is the value of q?
यदि 2 समीकरण $x^2 - px + 6 = 0$ का एक मूल है और द्विघात समीकरण $x^2 + 2px + q = 0$ के मूल बराबर है, तो q का मान क्या है?
(a) 25 (b) 36
(c) 12 (d) 16

31. If $a + b + c = 6$ and $ab + bc + ca = 5$, then $a^3+b^3 + c^3 - 3abc$ is equal to:
यदि $a + b + c = 6$ और $ab + bc + ca = 5$ है, तो $a^3+b^3 + c^3-3abc$ बराबर है:
(a) 116 (b) 126
(c) 98 (d) 108

32. If $\left\{\begin{matrix} a+b=48-ab \\ b+c=99-bc \\ c+a=3-ca \end{matrix}\right\}$ Where, a, b, c are positive no. find $7c - 15a + b = ?$

यदि $\left\{\begin{matrix} a+b=48-ab \\ b+c=99-bc \\ c+a=3-ca \end{matrix}\right\}$ जहाँ a, b, c धनात्मक संख्याएँ हैं, तो $7c - 15a + b = ?$ ज्ञात कीजिए
(a) 38 (b) 41
(c) 43 (d) 35

33. If $ab + bc + ca = 8$ and $a^2 + b^2 + c^2 = 20$, then a possible value of $\frac{1}{2}(a + b + c)[(a - b)^2 + (b - c)^2 + (c - a)^2]$ is:

यदि $ab + bc + ca = 8$ और $a^2 + b^2 + c^2 = 20$ है, तो $\frac{1}{2}(a + b + c)[(a - b)^2 + (b - c)^2 + (c - a)^2]$ का संभावित मान है:
(a) 72 (b) 56
(c) 84 (d) 80

34. If $\sqrt{(1-p^2)(1-q^2)} = \frac{\sqrt{3}}{2}$, then what is the value of $\sqrt{2p^2+2q^2+2pq} + \sqrt{2p^2+2q^2-2pq}$

यदि $\sqrt{(1-p^2)(1-q^2)} = \frac{\sqrt{3}}{2}$ है, तो $\sqrt{2p^2+2q^2+2pq} + \sqrt{2p^2+2q^2-2pq}$ का मान क्या है
(a) 2 (b) $\sqrt{2}$
(c) 1 (d) none of these

35. If $a^2 + b^2 + \frac{1}{a^2} + \frac{1}{b^2} = 4$, $a \neq 0$, $b \neq 0$, then the value of $a^4 + b^4$ is:

यदि $a^2 + b^2 + \frac{1}{a^2} + \frac{1}{b^2} = 4$, $a \neq 0$, $b \neq 0$, तो $a^4 + b^4$ का मान है।
(a) 32 (b) 256
(c) 64 (d) 2

36. Which of the following statement is correct?
निम्नलिखित में से कौन सा कथन सही है?

I. The value of $100^2 - 99^2 + 98^2 - 97^2 + 96^2 - 95^2 + 94^2 - 93^2 + \ldots\ldots\ldots + 2^2 + 1^2$ is 5050.
$100^2 - 99^2 + 98^2 - 97^2 + 96^2 - 95^2 + 94^2 - 93^2 + \ldots\ldots\ldots + 2^2 + 1^2$ का मान 5050 है।

II. If $8x + \frac{8}{x} = 16$ and $x < 0$, then the value of $x^{197} + x^{-197}$ is 2.

II. यदि $8x + \frac{8}{x} = 16$ और $x < 0$, तो $x^{197} + x^{-197}$ का मान 2 है।

(a) Only I/केवल I
(b) Both I and II/ I एक और II दोनों
(c) Neithern I nor II/ न तो I और न ही II
(d) Only II/ केवल II

37. What is the value of/का मूल्य है।

$$\left(k-\frac{1}{k}\right)\left(k^2+\frac{1}{k^2}\right)\left(k^4+\frac{1}{k^4}\right)\left(k^8+\frac{1}{k^8}\right)\left(k^{16}+\frac{1}{k^{16}}\right)\left(k^{32}+\frac{1}{k^{32}}\right)$$

(a) $\dfrac{k^{64}-\frac{1}{k^{64}}}{k+\frac{1}{k}}$ (b) $\dfrac{k^{32}-\frac{1}{k^{32}}}{k-\frac{1}{k}}$

(c) $\dfrac{k^{32}-\frac{1}{k^{32}}}{k+\frac{1}{k}}$ (d) $\dfrac{k^{32}+\frac{1}{k^{64}}}{k+\frac{1}{k}}$

38. If $x^4 + x^2y^2 + y^4 = \frac{21}{256}$ and $x^2 + xy + y^2 = \frac{3}{16}$, then $2(x^2 + y^2) = ?$

यदि $x^4 + x^2y^2 + y^4 = \frac{21}{256}$ और $x^2 + xy + y^2 = \frac{3}{16}$ है, तो $2(x^2 + y^2) = ?$

(a) $\frac{5}{16}$ (b) $\frac{5}{8}$

(c) $\frac{3}{8}$ (d) $\frac{3}{4}$

39. If $a + b = \frac{3}{4}$ and $ab = \frac{1}{8}$, them the value of $a^4 + b^4 + ab^3 + ba^3$ is:

यदि $a + b = \frac{3}{4}$ और $ab = \frac{1}{8}$ है, तो $a^4 + b^4 + ab^3 + ba^3$ का मान है:

(a) $\frac{27}{128}$ (b) $\frac{9}{256}$

(c) $\frac{27}{256}$ (d) $\frac{9}{64}$

40. If $x - y = \frac{7}{9}$ and $\frac{1}{x}-\frac{1}{y}=10\frac{1}{2}$, then what is the value of $(x^3 - y^3)$?

यदि $x - y = \frac{7}{9}$ और $\frac{1}{x}-\frac{1}{y}=10\frac{1}{2}$ है, तो $(x^3 - y^3)$ का मान क्या है?

(a) $\frac{26}{81}$ (b) $\frac{217}{729}$

(c) $\frac{455}{729}$ (d) $\frac{8}{27}$

41. If $x = \left(7+3\sqrt{5}\right)$, then find the value of $x^2 + \frac{1}{x^2}$.

यदि $x = \left(7+3\sqrt{5}\right)$ है, तो $x^2 + \frac{1}{x^2}$ का मान ज्ञात कीजिए।

(a) $\frac{580+315\sqrt{5}}{8}$ (b) $\frac{799+328\sqrt{5}}{8}$

(c) $\frac{799+315\sqrt{5}}{12}$ (d) $\frac{799+315\sqrt{5}}{8}$

42. If $x - y = 4$ and $x^3 - y^3 = 316$, $y > 0$ then the value of $x^4 - y^4$ is:

यदि $x - y = 4$ और $x^3 - y^3 = 316$, $y > 0$ तो $x^4 - y^4$ का मान है:

(a) 2482 (b) 2320

(c) 2500 (d) 2401

43. If $x = \sqrt{10}+\sqrt{11}$, $y = \sqrt{10}-\sqrt{11}$, then the value of $7x^2 - 50xy + 7y^2 = ?$

यदि $x = \sqrt{10}+\sqrt{11}$, $y = \sqrt{10}-\sqrt{11}$ है, तो $7x^2 - 50xy + 7y^2$ का मान क्या होगा?

(a) 344 (b) 704

(c) 1360 (d) 386

44. If $x+\frac{1}{x}=\sqrt{3}$, find

$x^{93} + x^{91} + x^{87} + x^{85} + x^{83} + x^{89} = ?$

यदि $x+\frac{1}{x}=\sqrt{3}$ है, तो $x^{93} + x^{91} + x^{87} + x^{85} + x^{83} + x^{89} = ?$

(a) 1 (b) 3

(c) 0 (d) –1

45. If $\frac{x+\sqrt{x^2-1}}{x-\sqrt{x^2-1}}+\frac{x-\sqrt{x^2-1}}{x+\sqrt{x^2-1}}=34$, then find the value of x (where $x < 0$)

यदि $\frac{x+\sqrt{x^2-1}}{x-\sqrt{x^2-1}}+\frac{x-\sqrt{x^2-1}}{x+\sqrt{x^2-1}}=34$, है, तो x (जहाँ $x < 0$) का मान है?

(a) ±1 (b) ±2

(c) ±3 (d) ±4

46. Simplify the following expression.

निम्नलिखित अभिव्यक्ति को सरल कीजिए।

$$\frac{5\left(a^6-b^6\right)^3+5\left(b^6-c^6\right)^3+5\left(c^6-a^6\right)^3}{2\left(a^3-b^3\right)^3+2\left(b^3-c^3\right)^3+2\left(c^3-a^3\right)^3}$$

(a) $\frac{5}{2}(a^3 + b^3)(b^3 + c^3)(c^3 + a^3)$

(b) $\frac{5}{2}(a^3 + b^3)(b^3 - c^3)(c^3 - a^3)$

(c) $\frac{5}{2}(a^3 - b^3)(b^3 + c^3)(c^3 + a^3)$

(d) $\frac{5}{2}(a^3 - b^3)(b^3 - c^3)(c^3 + a^3)$

47. If $p + q + r = 0$, then what is the simplified value of the expression

$$\left(\frac{p^2}{p^2-qr}+\frac{q^2}{q^2-pr}+\frac{r^2}{r^2-pq}\right)$$

यदि $p + q + r = 0$ है, तो व्यंजक $\left(\frac{p^2}{p^2-qr}+\frac{q^2}{q^2-pr}+\frac{r^2}{r^2-pq}\right)$ का सरलीकृत मान क्या है?

(a) 0 (b) 2

(c) 1 (d) – 1

48. If $\alpha + \beta + \gamma = 0$, then $\frac{3\beta^2+\alpha^2+\gamma^2}{2\beta^2-\alpha\gamma}=?$

यदि $\alpha + \beta + \gamma = 0$ है, तो $\frac{3\beta^2+\alpha^2+\gamma^2}{2\beta^2-\alpha\gamma}=?$

(a) $\frac{1}{2}$ (b) 1

(c) 2 (d) $\frac{1}{4}$

49. Which of the following statement is correct?
निम्नलिखित में से कौन सा कथन सही है?

I. If $x = 12$, $y = -2$ and $z = -10$, then $x^3 + y^3 + z^3 = 360$.
यदि $x = 12$, $y = -2$ और $z = -10$, तब $x^3 + y^3 + z^3 = 360$.

II. If $x + y = 48$ and $4xy = 128$, then $4x^2 + 4y^2 = 4480$.
यदि $x + y = 48$ और $4xy = 128$, तब $4x^2 + 4y^2 = 4480$.

(a) Neither I nor II
(b) Only I
(c) Both I and II
(d) Only II

50. If $\sqrt{\frac{a}{b}} = \frac{8}{3} - \sqrt{\frac{b}{a}}$ and $a - b = 10$, then the value of ab is :
यदि $\sqrt{\frac{a}{b}} = \frac{8}{3} - \sqrt{\frac{b}{a}}$ और $a - b = 10$ है, तो ab का मान है:

(a) $32\frac{1}{7}$
(b) $32\frac{3}{7}$
(c) $32\frac{4}{7}$
(d) $32\frac{2}{7}$

ANSWER KEY

1.	(b)	2.	(b)	3.	(c)	4.	(b)	5.	(a)	6.	(a)	7.	(b)	8.	(c)	9.	(b)	10.	(b)
11.	(d)	12.	(b)	13.	(b)	14.	(a)	15.	(a)	16.	(b)	17.	(a)	18.	(b)	19.	(c)	20.	(c)
21.	(d)	22.	(a)	23.	(c)	24.	(a)	25.	(a)	26.	(d)	27.	(c)	28.	(a)	29.	(d)	30.	(a)
31.	(b)	32.	(b)	33.	(a)	34.	(b)	35.	(d)	36.	(b)	37.	(a)	38.	(b)	39.	(c)	40.	(b)
41.	(d)	42.	(b)	43.	(a)	44.	(c)	45.	(c)	46.	(a)	47.	(b)	48.	(c)	49.	(a)	50.	(a)

SOLUTIONS

1. (b) $x^4 + \frac{1}{x^4} = 1154$, $x > 0$

$\Rightarrow x^2 + \frac{1}{x^2} = \sqrt{1154 + 2} = 34$

$\Rightarrow x + \frac{1}{x} = \sqrt{34 + 2} = \sqrt{36} = 6$ .....(i)

$\Rightarrow x - \frac{1}{x} = \sqrt{36 - 4} = \sqrt{32} = 4\sqrt{2}$ .....(ii)

Eqⁿ (i) + (ii)

$2x = 6 + 4\sqrt{2} \Rightarrow 2x - 4\sqrt{2} = 6$

$\therefore (2x - 4\sqrt{2})^2 = 6^2 = 36$

2. (b) $x^4 + \frac{1}{x^4} = 2207$

$\Rightarrow x^2 + \frac{1}{x^2} = \sqrt{2207 + 2} = \sqrt{2209} = 47$

$\Rightarrow x + \frac{1}{x} = \sqrt{47 + 2} = 7$

$\Rightarrow x^3 + \frac{1}{x^3} = 7^3 - 3 \times 7 = 322$

$\therefore x^7 + \frac{1}{x^7} = \left(x^3 + \frac{1}{x^3}\right)\left(x^4 + \frac{1}{x^4}\right) - \left(x + \frac{1}{x}\right)$

$= 322 \times 2207 - 7 = 710647$

SMART APPROACH:-

$x^4 + \frac{1}{x^4} = 2207$

$\Rightarrow x^2 + \frac{1}{x^2} = \sqrt{2207 + 2} = \sqrt{2209} = 47$

$\Rightarrow x + \frac{1}{x} = \sqrt{47 + 2} = 7$

If $x \pm \frac{1}{x} = k$ then unit digit of

$x^7 \pm \frac{1}{x^7}$ must be k.

Hence only option (b) satisfy the condition

3. (c) Given,

$\left(x - \frac{1}{x}\right) = 2\sqrt{2}$

$\Rightarrow x^3 - \frac{1}{x^3} = \left(2\sqrt{2}\right)^3 + 3 \times 2\sqrt{2} = 22\sqrt{2}$

$x + \frac{1}{x} = \sqrt{\left(2\sqrt{2}\right)^2 + 4} = 2\sqrt{3}$

$\Rightarrow x^3 + \frac{1}{x^3} = \left(2\sqrt{3}\right)^3 - 3 \times 2\sqrt{3}$

$= 24\sqrt{3} - 6\sqrt{3} = 18\sqrt{3}$

$\therefore x^6 - \frac{1}{x^6} = \left(x^3 - \frac{1}{x^3}\right) \times \left(x^3 + \frac{1}{x^3}\right)$

$= 22\sqrt{2} \times 18\sqrt{3} = 396\sqrt{6}$

Alternate Method:

$\left(x - \frac{1}{x}\right) = 2\sqrt{2}$

$\Rightarrow x + \frac{1}{x} = \sqrt{\left(2\sqrt{2}\right)^2 + 4} = 2\sqrt{3}$

$x^2 - \frac{1}{x^2} = 4\sqrt{6}$

$\therefore x^6 - \frac{1}{x^6} - 3 \times 4\sqrt{6} = (4\sqrt{6})^3$

$\Rightarrow x^6 - \frac{1}{x^6} = 384\sqrt{6} + 12\sqrt{6} = 396\sqrt{6}$

4. (b) $x = \frac{5 + \sqrt{21}}{2} = \frac{5}{2} + \frac{\sqrt{21}}{2}$

$\frac{1}{x} = \frac{5}{2} - \frac{\sqrt{21}}{2}$

$\therefore x + \frac{1}{x} = 5$

$\Rightarrow x^2 + \frac{1}{x^2} = 5^2 - 2 = 23$

$\Rightarrow x^4 + \frac{1}{x^4} = 23^2 - 2 = 527$

$\Rightarrow x^3 + \frac{1}{x^3} = 5^3 - 3 \times 5 = 110$

$\therefore x^5 + \frac{1}{x^5} = 110 \times 23 - 5 = 2525$

Now,

$x^{11} + x^7 - 2525x^6 - 527x^3$

$= x^{11} + x^7 - \left(x^5 + \frac{1}{x^5}\right)x^6 - \left(x^4 + \frac{1}{x^4}\right)x^3$

$= -\left(x + \frac{1}{x}\right) = -5$

5. (a) $\frac{7(2\sqrt{2} - x)}{x} = 7 + \frac{14}{x^2}$

$\Rightarrow \frac{(14\sqrt{2} - 7x)}{x} = \frac{(7x^2 + 14)}{x^2}$

$\Rightarrow (14\sqrt{2} - 7x) = 7x + \frac{14}{x}$

$\Rightarrow 14x + \frac{14}{x} = 14\sqrt{2}$

$\Rightarrow x + \frac{1}{x} = \sqrt{2}$

On squaring,

$x^2 + \frac{1}{x^2} = 0 \Rightarrow x^4 = -1$

$\therefore \sqrt{23x^{64} + 42x^{12} + 163}$

$= \sqrt{23 \times (x^4)^{16} + 42(x^4)^3 + 163}$

$= \sqrt{23 - 42 + 163} = \sqrt{144} = 12$

6. (a) $x^2 - 3\sqrt{7}x + 13 = 0$

Put, $x - \sqrt{7} = y$

$\Rightarrow x = (y + \sqrt{7})$

$(y + \sqrt{7})^2 - 3\sqrt{7}(y + \sqrt{7}) + 13 = 0$

$\Rightarrow y^2 + 7 + 2\sqrt{7}y - 3\sqrt{7}y - 21 + 13 = 0$

$\Rightarrow y^2 - \sqrt{7}y - 1 = 0$

$\Rightarrow y - \frac{1}{y} = \sqrt{7}$

$y^3 - \frac{1}{y^3} = 7\sqrt{7} + 3\sqrt{7} = 10\sqrt{7}$

$y + \frac{1}{y} = \sqrt{(\sqrt{7})^2 + 4} = \sqrt{11}$

$y^3 + \frac{1}{y^3} = 11\sqrt{11} - 3\sqrt{11} = 8\sqrt{11}$

$y^6 - \frac{1}{y^6} = 80\sqrt{77}$

7. (b) If $x = \frac{\sqrt{3} + 1}{\sqrt{3} - 1}$, $y = \frac{1}{x}$

By rationalization,

$x = \frac{\sqrt{3} + 1}{\sqrt{3} - 1} \times \frac{\sqrt{3} + 1}{\sqrt{3} + 1} = 2 + \sqrt{3}$

$y = \frac{1}{x} = \frac{1}{2 + \sqrt{3}} = 2 - \sqrt{3}$

$x + y = 4$ or $x + \frac{1}{x} = 4$ and $x - y = 2\sqrt{3}$

$x^3 + \frac{1}{x^3} = \left(x + \frac{1}{x}\right)^3 - 3\left(x + \frac{1}{x}\right)$

$= (4)^3 - 3 \times 4 = 52$

$\therefore x^3 - y^3 = (2\sqrt{3})^3 + 3 \times 2\sqrt{3}$

$= 24\sqrt{3} + 6\sqrt{3} = 30\sqrt{3}$

and, $xy = \left(2 + \sqrt{3}\right)\left(2 - \sqrt{3}\right) = 1$

$\therefore \frac{x^3 + \frac{1}{x^3} + 8xy}{x^3 - y^3}$

$= \frac{52 + 8}{30\sqrt{3}} = \frac{60}{30\sqrt{3}} = \frac{2}{\sqrt{3}} = \frac{2\sqrt{3}}{3}$

8. (c) $a + b + c = 4$, $a^2 + b^2 + c^2 = 14$

$\frac{1}{a} + \frac{1}{b} + \frac{1}{c} = 1$

$\Rightarrow \frac{bc + ac + ab}{abc} = 1$

$\Rightarrow ab + bc + ca = abc$

$(a + b + c)^2 = (4)^2$

$\Rightarrow a^2 + b^2 + c^2 + 2(ab + bc + ca) = 16$

$\Rightarrow 14 + 2abc = 16$

$\Rightarrow abc = 1$

9. (b) $x + y + z = -7$

Squaring both sides,

$49 = 133 + 2(xy + yz + zx)$

$\Rightarrow -84 = 2(x + y + z)y$

$\Rightarrow -42 = -7(y)$

$\Rightarrow y = 6$

$x + z = -6 - 7 = -13$

$x^2 + z^2 = 97$

Let, $x = -9$, $z = -4$

$\therefore z - x = -4 + 9 = 5$

10. (b) $a + b + c = 17$, $a^2 + b^2 + c^2 = 101$

$(a - b)^2 + (b - c)^2 + (c - a)^2$

$= 2[a^2 + b^2 + c^2 - ab - bc - ca]$

$(a + b + c)^2 = a^2 + b^2 + c^2 + 2(ab + bc + ca)$

$\Rightarrow 289 = 101 + 2(ab + bc + ca)$

$\Rightarrow 188 = 2(ab + bc + ca)$

$\Rightarrow ab + bc + ca = 94$

$\therefore (a - b)^2 + (b - c)^2 + (c - a)^2 = 2[101 - 94] = 2 \times 7 = 14$

11. (d) $(11x + 8)^3 + (2 - x)^3 - 27(2x - 5)^3 = 9(11x + 8)(x - 2)(2x - 5)$

We know that if $a + b + c = 0$ then $a^3 + b^3 + c^3 = 3abc$ or vice versa

$\Rightarrow (11x + 8)^3 + (2 - x)^3 - 3^3(2x - 5)^3 = -3 \times (11x + 8) \times (2 - x) \times 3(2x - 5)$

$a = 11x + 8, b = 2 - x$ & $c = -3(2x - 5)$

$\Rightarrow (11x + 8) + (2 - x) - 3(2x - 5) = 0$

$\Rightarrow 11x + 8 + 2 - x - 6x + 15 = 0$

$\Rightarrow 4x + 25 = 0$

$\Rightarrow x = \frac{-25}{4}$

$\therefore \sqrt{x^2 - 36} = \sqrt{\frac{625 - 576}{16}} = \sqrt{\frac{49}{16}} = \frac{7}{4}$

12. (b) $x^2 - 11x + 1 = 0$

$\Rightarrow x + \frac{1}{x} = 11$

$\Rightarrow x^2 + \frac{1}{x^2} = 121 - 2 = 119$

$\Rightarrow x^4 + \frac{1}{x^4} = 119^2 - 2 = 14161 - 2 = 14159$

$\Rightarrow x^8 + 1 = 14159x^4$

$\Rightarrow x^8 - 14159x^4 = -1$

$\Rightarrow x^8 - 14159x^4 + 11 = -1 + 11 = 10$

13. (b) ATQ,

$$\frac{2x}{x+2} + \frac{1}{2x+4} - \frac{12-x}{2x^2+4x} = a + \frac{b}{x}$$

$$\Rightarrow \frac{2x}{x+2} + \frac{1}{2(x+2)} - \frac{12-x}{2x(x+2)} = a + \frac{b}{x}$$

$$\Rightarrow \frac{4x^2 + x - 12 + x}{2x(x+2)} = \frac{ax+b}{x}$$

$$\Rightarrow \frac{4x^2 + 2x - 12}{2x(x+2)} = \frac{ax+b}{x}$$

$$\Rightarrow \frac{2x^2 + x - 6}{x+2} = ax + b$$

$\Rightarrow 2x^2 + x - 6 = ax^2 + bx + 2ax + 2b$

Coefficient $= 2x^2 = ax^2$

$\Rightarrow a = 2$

$-6 = 2b \Rightarrow b = -3$

$\therefore ab = 2 \times (-3) = -6$

Alternate Method:

Put $x = 1$

$$\Rightarrow \frac{2}{3} + \frac{1}{6} - \frac{11}{6} = a + b$$

$$\Rightarrow \frac{4 + 1 - 11}{6} = a + b$$

$\Rightarrow -1 = a + b$

Let, $a = 2, b = -3$

$\therefore ab = 2 \times (-3) = -6$

14. (a) $\frac{A}{L} + \frac{M}{B} = 1$

Let, $\frac{A}{L} = x$, $\frac{M}{B} = y$ and $\frac{C}{N} = z$

ATQ, $\frac{A}{L} + \frac{M}{B} = 1$

$x + y = 1, \; x = 1 - y$

$$\frac{1}{x} = \frac{1}{1-y} \quad \text{.....(i)}$$

Again, $\frac{B}{M} + \frac{N}{C} = 1$

$$\frac{1}{y} + \frac{1}{z} = 1, \quad \frac{1}{z} = 1 - \frac{1}{y} = \frac{y-1}{y}$$

$$z = \frac{y}{y-1} \quad \text{.....(ii)}$$

Put the value from equation (i) and (ii)

$$\frac{L}{A} + \frac{C}{N} = \frac{1}{x} + z = \frac{1}{1-y} + \frac{y}{y-1} = \frac{1-y}{1-y} = 1$$

SMART APPROACH:-

Let, $\frac{A}{L} = x, \frac{B}{M} = y, \frac{C}{N} = z$

Then, $x + \frac{1}{y} = 1, y + \frac{1}{z} = 1$

There is two equation and 3 variables

So, we can use value putting,

Let, $x = \frac{1}{2}$, $y = 2$ then $z = -1$

$$\frac{L}{A} + \frac{C}{N} = \frac{1}{x} + z = 2 - 1 = 1$$

15. (a) $\frac{x^6 - 24x^4 + 144x^2}{(x^2 + 4\sqrt{3}x + 12)(x - 2\sqrt{3})^2}$

$$= \frac{x^6 - 24x^4 + 144x^2}{(x^2 + 4\sqrt{3}x + 12)(x^2 + 12 - 4\sqrt{3}x)}$$

$$= \frac{x^2(x^4 - 24x^2 + 144)}{(x^2 + 12)^2 - (4\sqrt{3}x)^2}$$

$$= \frac{x^2(x^4 - 24x^2 + 144)}{x^4 + 144 + 24x^2 - 48x^2}$$

$$= \frac{x^2(x^4 - 24x^2 + 144)}{x^4 - 24x^2 + 144} = x^2$$

16. (b) $P = \frac{\sqrt{3q+2} + \sqrt{3q-2}}{\sqrt{3q+2} - \sqrt{3q-2}}$

Use CND Rule

$$\frac{P+1}{P-1} = \frac{\sqrt{3q+2}}{\sqrt{3q-2}}$$

Squaring both side

$$\Rightarrow \frac{P^2 + 1 + 2P}{P^2 + 1 - 2P} = \frac{3q+2}{3q-2}$$

Again use CND rule

$$\Rightarrow \frac{P^2 + 1}{2P} = \frac{3q}{2}$$

$\Rightarrow P^2 + 1 = 3pq$

$\Rightarrow P^2 - 3pq + 2 = 1$

17. (a) $a^4 + \frac{1}{a^4} = 50$

$$a^4 + \frac{1}{a^4} + 2 = 50 + 2$$

$$\left(a^2 + \frac{1}{a^2}\right)^2 = 52 \Rightarrow a^2 + \frac{1}{a^2} = 2\sqrt{13}$$

$$a^2 + \frac{1}{a^2} + 2 = 2\sqrt{13} + 2$$

$$\left(a + \frac{1}{a}\right)^2 = 2(\sqrt{13} + 1)$$

$a+\frac{1}{a}=\sqrt{2\left(\sqrt{13}+1\right)}$

Let, $x=\sqrt{2(\sqrt{13}+1)}$

$a^3+\frac{1}{a^3}=x^3-3x=x(x^2-3)$

$=\sqrt{2(\sqrt{13}+1)}\left[2(\sqrt{13}+1)-3\right]$

$=\sqrt{2(\sqrt{13}+1)}\ \left[2\sqrt{13}+2-3\right]$

$=\sqrt{2(\sqrt{13}+1)}\ \left[2\sqrt{13}-1\right]$

18. (b) ATQ,

$\frac{6x}{2x^2+5x-2}=1$

$\Rightarrow 6x=2x^2+5x-2$

$\Rightarrow x=2x^2-2 \Rightarrow x-\frac{1}{x}=\frac{1}{2}$

If $x-\frac{1}{x}=k$, then $x+\frac{1}{x}=\sqrt{k^2+4}$

$x+\frac{1}{x}=\sqrt{\left(\frac{1}{2}\right)^2+4}=\frac{\sqrt{17}}{2}$

If $x+\frac{1}{x}=k$,

then $x^3+\frac{1}{x^3}=k^3-3k$

$x^3+\frac{1}{x^3}=\left(\frac{\sqrt{17}}{2}\right)^3-3\times\frac{\sqrt{17}}{2}$

$=\frac{17\sqrt{17}}{8}-\frac{3\sqrt{17}}{2}=\frac{5\sqrt{17}}{8}$

19. (c) ATQ,

$x+\frac{1}{x}=\sqrt{6}$

$\Rightarrow x^2+\frac{1}{x^2}=\left(\sqrt{6}\right)^2-2=4$

$\Rightarrow x^4+\frac{1}{x^4}=4^2-2=14$

$\Rightarrow x-\frac{1}{x}=\sqrt{\left(\sqrt{6}\right)^2-4}=\sqrt{2}$

$\Rightarrow x^8-\frac{1}{x^8}=\left(x^4+\frac{1}{x^4}\right)\times\left(x^4-\frac{1}{x^4}\right)$

$=\left(x^4+\frac{1}{x^4}\right)\times\left(x^2+\frac{1}{x^2}\right)\left(x+\frac{1}{x}\right)\left(x-\frac{1}{x}\right)$

$=14\times4\times\sqrt{6}\times\sqrt{2}=112\sqrt{3}$

20. (c) I. $100^2-99^2+98^2-97^2+96^2-95^2+94^2-93^2+\ldots\ldots\ldots+22^2-21^2$

$=(100-99)(100+99)+(98-97)(98+97)+\ldots\ldots+(22-21)(22+21)=100+99+98+97+\ldots\ldots+22+21$

$=\frac{100\times101}{2}-\frac{20\times21}{2}=5050-210=4840.$

$\therefore$ It is true

II. $\left(k^2+\frac{1}{k^2}\right)\left(k-\frac{1}{k}\right)\left(k^4+\frac{1}{k^4}\right)\left(k+\frac{1}{k}\right)\left(k^4-\frac{1}{k^4}\right)$

$=\left(k^2+\frac{1}{k^2}\right)\left(k^2-\frac{1}{k^2}\right)\left(k^4+\frac{1}{k^4}\right)\left(k^4-\frac{1}{k^4}\right)$

$=\left(k^4-\frac{1}{k^4}\right)\left(k^8-\frac{1}{k^8}\right)\neq\left(k^{16}-\frac{1}{k^{16}}\right)$

$\therefore$ It is not true

So, only option (c) is correct.

21. (d) $3a+2b=27$, $(27a^3+8b^3)=1458$, $2ab=?$

Cube both sides,

$27a^3+8b^3+3\times3a\times2b\times(3a+2b)=27^3$

$\Rightarrow 27a^3+8b^3+18ab\times27=27^3$

$\Rightarrow 1458+18ab\times27=27^3$

$\Rightarrow 54+18ab=27^2$

$\Rightarrow 6+2ab=27\times3$

$\Rightarrow 2ab=27\times3-6 \Rightarrow 2ab=75$

22. (a) $8a^3-b^3=(2a)^3-(b)^3$

$\Rightarrow 999=(2a-b)[(2a-b)^2+6ab]$

$\Rightarrow 999=3[9+6ab]$

$\Rightarrow 333-9=6ab$

$\Rightarrow ab=\frac{324}{6}=54$

$\because (2a+b)^2=(2a-b)^2+4(2a)(b)$

$=9+432=441$

$\Rightarrow 2a+b=21$

$\therefore 4a^2-b^2=(2a+b)(2a-b)$

$=21\times3=63$

Alternate Method:-

Put $a=6$, $b=9$

$\Rightarrow 2a-b=3$

$\Rightarrow 2\times6-9=3$ (satisfied)

$8a^3-b^3=999$

$\Rightarrow 8(6)^3-(9)^3=1728-729=999$ (satisfied)

$\therefore 4a^2-b^2=144-81=63$

23. (c) $(4x+2y)^3+(4x-2y)^3=16(Ax^3+Bxy^2)$ ....(i)

$[(A+B)^3+(A-B)^3=A^3+B^3+3A^2B+3AB^2+A^3-B^3-3A^2B+3AB^2$

$=2A^3+6AB^2]$

$\therefore (4x+2y)^3+(4x-2y)^3=2\times(4x)^3+6\times4x\times(2y)^2$

$=128x^3+96xy^2$

$=16(8x^3+6xy^2)$ ....(ii)

Comparing equation (i) and (ii),

$A=8$, $B=6$

$\therefore \frac{1}{2}\left(\sqrt{A^2+B^2}\right)=\frac{1}{2}\left(\sqrt{64+36}\right)=\frac{10}{2}=5$

24. (a) Given,

$a+b+c=20$

$a^2+b^2+c^2=152$

Then,

$(a+b+c)^2=a^2+b^2+c^2+2(ab+bc+ca)$

$\Rightarrow 400=152+2(ab+bc+ca)$

$\Rightarrow 248=2(ab+bc+ca)$

$\Rightarrow 124=ab+bc+ca$

$\therefore a^3+b^3+c^3-3abc=(a+b+c)[a^2+b^2+c^2-(ab+bc+ca)]$

$=20[152-124]$

$=20[28]=560$

Alternate Method:

Here is three variable and two equation, So we can put the value of one variable 0.

Let, $c=0$

$a+b=20$, $a^2+b^2=152$, $a^3+b^3=?$

$(a + b)^2 = a^2 + b^2 + 2ab$

$\Rightarrow 400 = 152 + 2ab$

$\Rightarrow ab = \frac{248}{2} = 124$

$a^3 + b^3 = (a + b)[(a + b)^2 - 3ab] = 20[20^2 - 3 \times 124] = 560$

Using DS, $2(4 - 3) = 2$

Option (a) 560 has DS = '2'

SMART APPROACH:-

$a^3 + b^3 + c^3 - 3abc = \frac{a + b + c}{2}$

$[3(a^2 + b^2 + c^2) - (a + b + c)^2]$

$= \frac{20}{2}[3 \times 152 - 20^2]$

Using DS

$= 1[6 - 4] = 2$

$= 10 \times [456 - 400] = 560$

From options 560 has DS = 2

25. (a) $x = \sqrt{1+\frac{\sqrt{3}}{2}} - \sqrt{1-\frac{\sqrt{3}}{2}}$

Square both side

$x^2 = 1+\frac{\sqrt{3}}{2}+1-\frac{\sqrt{3}}{2}-2\sqrt{\left(1-\frac{\sqrt{3}}{2}\right)\left(1+\frac{\sqrt{3}}{2}\right)}$

$\Rightarrow x^2 = 2-2\sqrt{1-\frac{3}{4}}$

$\Rightarrow x^2 = 2 - 2 \times \frac{1}{2}$

$\Rightarrow x^2 = 1 \Rightarrow x = 1$

$\therefore \frac{\sqrt{2}-x}{\sqrt{2}+x} \times \frac{\sqrt{2}-x}{\sqrt{2}-x} = \frac{(\sqrt{2}-x)^2}{2-x^2}$

$= 2 + 1 - 2\sqrt{2}$

$= 3 - 2.82 \sim 0.17$

26. (d) $\frac{3\sqrt{5}-5}{3\sqrt{5}+5} = a + b\sqrt{5}$

$\Rightarrow \frac{3\sqrt{5}-5}{3\sqrt{5}+5} \times \frac{3\sqrt{5}-5}{3\sqrt{5}-5} = a + b\sqrt{5}$

$\Rightarrow \frac{45+25-30\sqrt{5}}{45-25} = a + b\sqrt{5}$

$\Rightarrow \frac{70-30\sqrt{5}}{20} = a + b\sqrt{5}$

$\Rightarrow \frac{70}{20} - \frac{30\sqrt{5}}{20} = a + b\sqrt{5}$

$\Rightarrow \frac{7}{2} - \frac{3}{2}\sqrt{5} = a + b\sqrt{5}$

$\therefore b = -\frac{3}{2}$

27. (c) $\left[(x-y)+\frac{y^2}{x+y}\right] \div \left[x^2+y^2+\frac{y^4}{x^2-y^2}\right] \times \frac{y^2}{x-y}$

$= \left[\frac{x^2-y^2+y^2}{x+y} \div \frac{x^4-y^4+y^4}{x^2-y^2}\right] \times \frac{y^2}{x-y}$

$= \left[\frac{x^2}{x+y} \times \frac{x^2-y^2}{x^4}\right] \times \frac{y^2}{x-y} = \frac{x-y}{x^2} \times \frac{y^2}{x-y} = \frac{y^2}{x^2}$

28. (a) $a^3 - b^3 = 625$, $a^2 - b^2 = 25$ and $a + b = 5$

We know that,

$a^2 - b^2 = (a + b)(a - b)$

$\Rightarrow 5(a - b) = 25 \Rightarrow (a - b) = 5$

We know that,

$a^3 - b^3 = (a - b)(a^2 + ab + b^2)$

$\Rightarrow (a - b)(a^2 + ab + b^2) = 625$

$\Rightarrow a^2 + ab + b^2 = \frac{625}{5} = 125$

29. (d) Let, $3a = 4b = 6c = k$

$\frac{3a}{12} = \frac{4b}{12} = \frac{6c}{12} = k$

$\Rightarrow a = 4k, b = 3k, c = 2k$

ATQ,

$4k + 3k + 2k = 27\sqrt{29}$

$\Rightarrow k = 3\sqrt{29}$

$\Rightarrow a = 12\sqrt{29}, b = 9\sqrt{29}, c = 6\sqrt{29}$

$\therefore \sqrt{a^2+b^2+c^2}$

$= \sqrt{\left(12\sqrt{29}\right)^2+\left(9\sqrt{29}\right)^2+\left(6\sqrt{29}\right)^2}$

$= \sqrt{29 \times (144 + 81 + 36)}$

$= \sqrt{29 \times 9 \times 29} = 87$

30. (a) $x^2 - px + 6 = 0$

$\Rightarrow 2^2 - p \times 2 + 6 = 0$

$\Rightarrow 4 + 6 - 2p = 0$

$\Rightarrow 2p = 10 \Rightarrow p = 5$

ATQ,

When roots are same

$\Rightarrow b^2 - 4ac = 0$

$\Rightarrow x^2 + 2px + q = 0$ .....(i)

putting value of p in equation (i)

Now, $x^2 + 10x + q = 0$

when roots are same. we know that,

$\Rightarrow b^2 - 4ac = 0$

$\Rightarrow 100 - 4 \times 1 \times q = 0 \Rightarrow 4q = 100$

$\Rightarrow q = 25$

31. (b) $a + b + c = 6$

$ab + bc + ca = 5$

Two equations and 3 variables so we can assume one variable zero,

so let, $c = 0$, $a + b = 6$, $ab = 5$

$a^2 + b^2 = (a+b)^2 - 2ab = 6^2 - 10 = 26$

Now, $a^3 + b^3 + c^3 - 3abc$

$c = 0$

then,

$a^3+b^3 = (a + b)(a^2 + b^2 - ab)$

$= (6)(26 - 5) = 6 \times 21 = 126$

Alternate Method:-

$a + b + c = 6$

$ab + bc + ca = 5$

$a^2 + b^2 + c^2 = (a + b + c)^2 - 2(ab + bc + ca) = 36 - 10 = 26$

$a^3 + b^3 + c^3 - 3abc = (a + b + c)[(a^2 + b^2 + c^2 - (ab + bc + ca)]$

$a^3 + b^3 + c^3 - 3abc = 6(26 - 5)$

$= 6 \times 21 = 126$

32. (b) $a + b = 48 - ab$

$\Rightarrow a + b + ab = 48$

$\Rightarrow 1 + a + b + ab = 48 + 1$

$\Rightarrow (1 + a)(1 + b) = 49$ .....(i)

Similarly,

$(1 + b)(1 + c) = 100$ .....(ii)

$(1 + c)(1 + a) = 4$ .....(iii)

Multiply equ. (i), (ii) and (iii)

$(1 + a)^2 (1 + b)^2 (1 + c)^2 = 49 \times 4 \times 100$

$\Rightarrow (1 + a)(1 + b)(1 + c) = 140$ ....(iv)

$(1 + a) = \frac{140}{100} = \frac{7}{5} \Rightarrow a = \frac{7}{5} - 1 = \frac{2}{5}$

$(1 + b) = \frac{140}{4} = 35 \Rightarrow b = 35 - 1 = 34$

$(1 + c) = \frac{140}{49} = \frac{20}{7} \Rightarrow c = \frac{20}{7} - 1 = \frac{13}{7}$

$\therefore 7c - 15a + b = \frac{7 \times 13}{7} - \frac{15 \times 2}{5} + 34$

$= 13 - 6 + 34 = 41$

33. (a) $ab + bc + ca = 8$

$a^2 + b^2 + c^2 = 20$

When there are 2 equations and 3 variables, so we can assume 1 variable zero

let, $c = 0$

$ab = 8 \Rightarrow a^2 + b^2 = 20$

let, $a = 4, b = 2$

$= \frac{1}{2}(a + b + c)[(a - b)^2 + (b - c)^2 + (c - a)^2]$

$= \frac{1}{2}(4 + 2 + 0)[(4 - 2)^2 + (2 - 0)^2 + (0 - 4)^2]$

$= (3)(4 + 4 + 16) = 72$

Alternate Method:-

$ab + bc + ca = 8$

$a^2 + b^2 + c^2 = 20$

$\therefore (a + b + c)^2 = a^2 + b^2 + c^2 + 2(ab + bc + ca) = 20 + 2(8)$

$\Rightarrow (a + b + c)^2 = 36$

$\Rightarrow a + b + c = 6$

Now,

$\frac{1}{2}(a + b + c)[(a - b)^2 + (b - c)^2 + (c - a)^2]$

$= \frac{1}{2}(a + b + c)[2(a^2 + b^2 + c^2) - 2(ab + bc + ca)]$

$= \frac{1}{2} \times 6\,(2 \times 20 - 2 \times 8) = 3 \times 24 = 72$

34. (b) $\sqrt{(1 - p^2)(1 - q^2)} = \frac{\sqrt{3}}{2}$

Squaring both sides

$(1 - p^2)(1 - q^2) = \frac{3}{4}$

Put $q = 0$

$\Rightarrow 1 - p^2 = \frac{3}{4} \Rightarrow p = \frac{1}{2}$

$\therefore \sqrt{2p^2 + 2q^2 + 2pq} + \sqrt{2p^2 + 2q^2 - 2pq}$

$= \sqrt{2 \times \frac{1}{4} + 0 + 0} + \sqrt{2 \times \frac{1}{4} + 0 - 0} = \frac{1}{\sqrt{2}} + \frac{1}{\sqrt{2}} = \frac{2}{\sqrt{2}} = \sqrt{2}$

35. (d) Put $a, b = 1$

$1^2 + 1^2 + \frac{1}{1^2} + \frac{1}{1^2} = 4$ (Satisfied)

$\therefore a^4 + b^4 = 1^4 + 1^4 = 1 + 1 = 2$

Alternate Method:-

$a^2 + b^2 + \frac{a^2 + b^2}{a^2 b^2} = 4$

$\Rightarrow a^2 b^2 (a^2 + b^2) + (a^2 + b^2) = 4a^2 b^2$

$\Rightarrow (a^2 + b^2)(a^2 b^2 + 1) = 4a^2 b^2$

$\Rightarrow a^2 + \frac{1}{a^2} + b^2 + \frac{1}{b^2} = 4$

$\Rightarrow a^2 + \frac{1}{a^2} - 2 + b^2 + \frac{1}{b^2} - 2 = 0$

$\Rightarrow \left(a - \frac{1}{a}\right)^2 + \left(b - \frac{1}{b}\right)^2 = 0$

$\Rightarrow a - \frac{1}{a} = b - \frac{1}{b} = 0$

$\therefore a = \frac{1}{a}, b = \frac{1}{b}$

$a^2 = 1, b^2 = 1$

Now, $a^4 + b^4 = (1)^2 + (1)^2 = 2$

36. (b) I $\rightarrow 100^2 - 99^2 + 98^2 - 97^2 + 96^2 - 95^2 + \ldots\ldots + 2^2 - 1^2$

Since $a^2 - b^2 = (a - b)(a + b)$

$\therefore (100 - 99)(100 + 99) + (98 - 97)(98 + 97) + \ldots\ldots (2 - 1)(2 + 1)$

$\Rightarrow 199, 195, 191, \ldots\ldots 7, 3$

No. of terms $= \frac{3 - 199}{-4} + 1 = 50$

$\therefore$ Sum $= \frac{n}{2}[2a + (n - 1)d]$

$= 25\,[2 \times 199 + 49\,(-4)]$

$= 25\,(398 - 196) = 5050$

$\therefore$ Statement I is true

II $\rightarrow 8x + \frac{8}{x} = 16$

$8\left(x + \frac{1}{x}\right) = 16 \Rightarrow x + \frac{1}{x} = 2 \Rightarrow x = 1$

$\therefore x^{197} + x^{-197} = x^{197} + \frac{1}{x^{197}} \Rightarrow 1 + 1 = 2$

$\therefore$ Statement II is true

37. (a) Multiply and divide by $\left(K + \frac{1}{K}\right)$

$$\Rightarrow \frac{\left(K + \frac{1}{K}\right)\left(K - \frac{1}{K}\right)\left(K^2 + \frac{1}{K^2}\right)\left(K^4 + \frac{1}{K^4}\right)\left(K^8 + \frac{1}{K^8}\right)\left(K^{16} + \frac{1}{K^{16}}\right)\left(K^{32} + \frac{1}{K^{32}}\right)}{\left(K + \frac{1}{K}\right)\left(K^2 - \frac{1}{K^2}\right)\left(K^2 + \frac{1}{K^2}\right)\left(K^4 + \frac{1}{K^4}\right)\left(K^8 + \frac{1}{K^8}\right)\left(K^{16} + \frac{1}{K^{16}}\right)\left(K^{32} + \frac{1}{K^{32}}\right)}{\left(K + \frac{1}{K}\right)}$$

$$\Rightarrow \frac{\left(K^4-\frac{1}{K^4}\right)\left(K^4+\frac{1}{K^4}\right)\left(K^8+\frac{1}{K^8}\right)\left(K^{16}+\frac{1}{K^{16}}\right)\left(K^{32}+\frac{1}{K^{32}}\right)}{\left(K+\frac{1}{K}\right)}$$

$$\Rightarrow \frac{\left(K^8-\frac{1}{K^8}\right)\left(K^8+\frac{1}{K^8}\right)\left(K^{16}+\frac{1}{K^{16}}\right)\left(K^{32}+\frac{1}{K^{32}}\right)}{\left(K+\frac{1}{K}\right)}$$

$$\Rightarrow \frac{\left(K^{16}-\frac{1}{K^{16}}\right)\left(K^{16}+\frac{1}{K^{16}}\right)\left(K^{32}+\frac{1}{K^{32}}\right)}{\left(K+\frac{1}{K}\right)}$$

$$\Rightarrow \frac{\left(K^{32}-\frac{1}{K^{32}}\right)\left(K^{32}+\frac{1}{K^{32}}\right)}{\left(K+\frac{1}{K}\right)} \Rightarrow \frac{K^{64}-\frac{1}{K^{64}}}{K+\frac{1}{K}}$$

38. (b) ATQ,

$$x^2 + xy + y^2 = \frac{3}{16}$$

$$\Rightarrow x^2 + y^2 = \frac{3}{16} - xy$$

Square both side

$$\Rightarrow x^4 + y^4 + 2x^2y^2 = \frac{9}{256} + x^2y^2 - \frac{3}{8}xy$$

$$\Rightarrow x^4 + y^4 + x^2y^2 = \frac{9}{256} - \frac{3}{8}xy$$

$$\Rightarrow \frac{21}{256} - \frac{9}{256} = -\frac{3}{8}xy$$

$$\Rightarrow \frac{12}{256} = \frac{-3}{8} \times xy = -\frac{1}{8} = xy$$

$$\therefore x^2 + y^2 + xy = \frac{3}{16}$$

$$\Rightarrow x^2 + y^2 - \frac{1}{8} = \frac{3}{16}$$

$$\Rightarrow x^2 + y^2 = \frac{5}{16}$$

Multiple by 2 both side

$$2(x^2 + y^2) = \frac{5}{8}$$

Alternate Method:-

Given,

$$x^4 + y^4 + x^2y^2 = \frac{21}{256}$$

$$x^2 + xy + y^2 = \frac{3}{16} \quad \text{.... (i)}$$

$$x^4 + y^4 + x^2y^2 = (x^2 + xy + y^2) \times (x^2 - xy + y^2)$$

$$\therefore x^2 - xy + y^2 = \frac{\frac{21}{256}}{\frac{3}{16}} = \frac{7}{16} \quad \text{.....(ii)}$$

Add eq. (i) + (ii)

$$2(x^2 + y^2) = \frac{3}{16} + \frac{7}{16} = \frac{5}{8}$$

39. (c) ATQ,

$$a + b = \frac{3}{4}$$

Square both sides

$$\Rightarrow a^2 + b^2 + 2ab = \frac{9}{16}$$

$$\Rightarrow a^2 + b^2 = \frac{5}{16}$$

Again square both sides

$$\Rightarrow a^4 + b^4 + 2a^2b^2 = \frac{25}{256}$$

$$\Rightarrow a^4 + b^4 = \frac{25}{256} - \frac{1}{32}$$

$$\Rightarrow a^4 + b^4 = \frac{17}{256}$$

$$\therefore a^4 + b^4 + ab^3 + ba^3$$

$$= a^4 + b^4 + ab(b^2 + a^2)$$

$$\frac{17}{256} + \frac{1}{8}\left(\frac{5}{16}\right) = \frac{17}{256} + \frac{5}{128} = \frac{27}{256}$$

40. (b) ATQ,

$$\frac{1}{x} - \frac{1}{y} = 10\frac{1}{2}$$

$$\Rightarrow \frac{-(x-y)}{xy} = \frac{21}{2}$$

$$\Rightarrow xy = -\frac{2}{27}$$

Given, $x - y = \frac{7}{9}$

Square both sides

$$\Rightarrow x^2 + y^2 - 2xy = \frac{49}{81}$$

$$\Rightarrow x^2 + y^2 = \frac{49}{81} - \frac{4}{27}$$

$$\Rightarrow x^2 + y^2 = \frac{49-12}{81} = \frac{37}{81}$$

$$\therefore x^3 - y^3 = (x-y)(x^2 + y^2 + xy)$$

$$= \frac{7}{9}\left(\frac{37}{81} - \frac{2}{27}\right) = \frac{7}{9}\left(\frac{37-6}{81}\right) = \frac{7}{9} \times \frac{31}{81} = \frac{217}{729}$$

41. (d) $x = 7 + 3\sqrt{5}$

then, $\frac{1}{x} = \frac{1}{7+3\sqrt{5}} \times \frac{7-3\sqrt{5}}{7-3\sqrt{5}}$

$$\Rightarrow \frac{1}{x} = \frac{7-3\sqrt{5}}{49-45} \Rightarrow \frac{1}{x} = \frac{7-3\sqrt{5}}{4}$$

So, $x + \frac{1}{x} = 7 + 3\sqrt{5} + \frac{7-3\sqrt{5}}{4}$

$$\Rightarrow x + \frac{1}{x} = \frac{28+12\sqrt{5}+7-3\sqrt{5}}{4}$$

On squaring both side,

$$\left(x + \frac{1}{x}\right)^2 = \left(\frac{35+9\sqrt{5}}{4}\right)^2$$

$$\Rightarrow x^2 + \frac{1}{x^2} = \left(\frac{35+9\sqrt{5}}{4}\right)^2 - 2$$

$$= \frac{1225+405+630\sqrt{5}}{16} - 2$$

$$= \frac{1630+630\sqrt{5}}{16} - 2$$

$$= \frac{1630+630\sqrt{5}-32}{16}$$

$$= \frac{1598+630\sqrt{5}}{16} = \frac{799+315\sqrt{5}}{8}$$

42. (b) ATQ,

$x^3 - y^3 = (x-y)[(x-y)^2 + 3xy]$

$316 = 4[16 + 3xy]$

$\Rightarrow 79 = 16 + 3xy$

$\Rightarrow xy = 21$

$(x-y)^2 = (x+y)^2 - 4xy$

$16 = (x+y)^2 - 84$

$\Rightarrow x + y = 10$

$(x+y)^2 = x^2 + y^2 + 2xy$

$100 = x^2 + y^2 + 42$

$\Rightarrow x^2 + y^2 = 58$

$x^4 - y^4 = (x+y)(x-y)(x^2+y^2)$

$= 10 \times 4 \times 58 = 2320$

Alternate Method:-

Put the value of $x = 7$ and $y = 3$

$x - y = 4$

$7 - 3 = 4$ (satisfied)

$x^3 - y^3 = 316$

$7^3 - 3^3 = 316$

$316 = 316$ (Satisfied)

then, $x^4 - y^4 = 7^4 - 3^4 = 2401 - 81 = 2320$

43. (a) $x = \sqrt{10} + \sqrt{11}$

$y = \sqrt{10} - \sqrt{11}$

$xy = \left(\sqrt{10}\right)^2 - \left(\sqrt{11}\right)^2$

$xy = -1$

$x = \frac{-1}{y}$

$7x^2 - 50xy + 7y^2$

$\Rightarrow 7(x^2+y^2) - 50(-1) \Rightarrow 7\left(x^2 + \frac{1}{x^2}\right) + 50$

From eqn (i) & (ii)

$x - y = x + \frac{1}{x} = \sqrt{10} + \sqrt{11} - \sqrt{10} + \sqrt{11} = 2\sqrt{11}$

$x^2 + \frac{1}{x^2} = \left(2\sqrt{11}\right)^2 - 2 = 42$

$\therefore 7\left(x^2 + \frac{1}{x^2}\right) + 50 = 7 \times 42 + 50 = 344$

44. (c) $x + \frac{1}{x} = \sqrt{3}$

$x^6 + 1 = 0$

$\therefore x^{93} + x^{91} + x^{87} + x^{85} + x^{83} + x^{89}$

$= x^{87}(x^6+1) + x^{85}(x^6+1) + x^{83}(x^6+1) = 0$

45. (c) Given,

$$\frac{x+\sqrt{x^2-1}}{x-\sqrt{x^2-1}} + \frac{x-\sqrt{x^2-1}}{x+\sqrt{x^2-1}} = 34$$

$$\Rightarrow \frac{\left(x+\sqrt{x^2-1}\right)^2 + \left(x-\sqrt{x^2-1}\right)^2}{\left(x-\sqrt{x^2-1}\right)\left(x+\sqrt{x^2-1}\right)} = 34$$

$$\Rightarrow \frac{\left(x^2+x^2-1+x^2+x^2-1\right)}{x^2-x^2+1} = 34$$

$\Rightarrow 4x^2 - 2 = 34$

$\Rightarrow 4x^2 = 36$

$\Rightarrow x^2 = 9 \Rightarrow x = \pm 3$

46. (a) Given,

$$\frac{5\left(a^6-b^6\right)^3 + 5\left(b^6-c^6\right)^3 + 5\left(c^6-a^6\right)^3}{2\left(a^3-b^3\right)^3 + 2\left(b^3-c^3\right)^3 + 2\left(c^3-a^3\right)^3}$$

We know that, if $x + y + z = 0$

Then,

$x^3 + y^3 + z^3 = 3xyz$

$$\Rightarrow \frac{5\times 3\times\left(a^3-b^3\right)\left(a^3+b^3\right)\left(b^3-c^3\right)\left(b^3+c^3\right)\left(c^3+a^3\right)\left(c^3-a^3\right)}{2\times 3\times\left(a^3-b^3\right)\left(b^3-c^3\right)\left(c^3-a^3\right)}$$

$$= \frac{5}{2}\left(a^3+b^3\right)\left(b^3+c^3\right)\left(c^3+a^3\right)$$

47. (b) $(p + q + r) = 0$

Put $p = 1, q = -1, r = 0$

$$\therefore \left(\frac{p^2}{p^2-qr}\right) + \left(\frac{q^2}{q^2-pr}\right) + \left(\frac{r^2}{r^2-pq}\right)$$

$= 1 + 1 + 0 = 2$

48. (c) Put $\alpha = \beta = 1$; $\gamma = -2$

$$\therefore \frac{3\beta^2+\alpha^2+\gamma^2}{2\beta^2-\alpha\gamma}$$

$$= \frac{3\times(1)^2+(1)^2+(-2)^2}{2\times(1)^2-1\times(-2)} = 2$$

49. (a) I. $x + y + z = 12 - 2 - 10 = 0$

$\Rightarrow x^3 + y^3 + z^3 = 3xyz$

$= 3 \times 12 \times (-2)(-10) = 720$

Statement I is not true

II. $4(x^2 + y^2) = 4[(x+y)^2 - 2xy]$

$= 4[(48)^2 - 64] = 8960$

Statement II is also not true

Both statements are false

50. (a) $\sqrt{\frac{a}{b}} + \sqrt{\frac{b}{a}} = \frac{8}{3}$

$$\Rightarrow \frac{a}{b} + \frac{b}{a} = \frac{64}{9} - 2 \Rightarrow \frac{a^2+b^2}{ab} = \frac{46}{9}$$

Now, $(a - b) = 10$

Squaring both side

$\Rightarrow a^2 + b^2 - 2ab = 100$

$$\Rightarrow \frac{46}{9}ab - 2ab = 100$$

$$\Rightarrow 28ab = 900 \Rightarrow ab = \frac{225}{7} = 32\frac{1}{7}$$